Auditing the Winner: An Upper Bound on What Critic Disagreement Adds Beyond the Selector on SWE-Agent Trajectories

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pre-registration, amendments, code, data
github.com/morantejr/silentfail-bench

The question

A coding agent tries a task n times; a scoring model picks the best attempt and commits it. **Silent failure:** the agent says it is done, the scorer agrees, the project's tests fail.

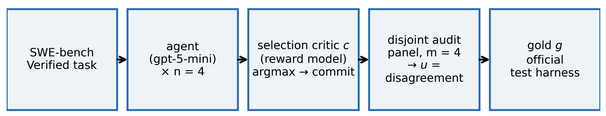
35% and 45%

of committed trajectories were silent failures
(SWE-bench Verified, 34 rows; Multilingual, 77 rows)

The proposed safeguard: a **disjoint panel of critics**; escalate when they **disagree**. Audit signals are almost always evaluated on candidate *pools*. A deployed auditor runs *after* selection. We evaluated it where it would run: on the population the selector actually commits, against the selector's own score, with executed gold.

Setup (pre-registered throughout)

- `gpt-5-mini` agent, 40-step limit; best-of-4 by a reward model (`Skywork-Reward-V2-Qwen3-4B`)
- Audit panel of four chat reward models, checkpoint-disjoint from the selector: `GRM-gemma2-2B`, `GRM-Llama3.2-3B`, `Skywork-Reward-V2-Llama-3.2-1B`, `Skywork-Reward-V2-Qwen3-1.7B`
- **Rows:** 41 Verified instances → 34 committed trajectories (7 excluded: empty patch or ungradable); 114 Multilingual → 77 (30 empty patches, 7 harness errors). The population is what the selector committed and the agent declared complete.
- Gold: trajectory-level success under the official `swebench` 5.x harness; positive class = failure
- **Glossary:** u = how much the four critics disagree (SD of their rank-normalised scores, ranks fit on training repositories only); $-c$ = the selector's own score with the sign flipped so higher means "more likely to fail"; LORO = leave-one-repository-out
- Endpoint: out-of-fold ΔAUROC of $(-c + u)$ over $-c$ alone; repository-clustered bootstrap



estimand: does u rank silent failures ($g = 0$ among committed, agent-declared-done trajectories) beyond c?

Takeaway for anyone building review layers for AI output

Evaluate an auditor on the outputs that ship, against the score you already have, with executed ground truth. Agreement among models is not a check; each model's own judgment is worth something, their disagreement is not. The thing that caught every failure here was running the real tests.

Result: a bound, not an absence

Disagreement alone ranks failures at **0.508** and **0.565** AUROC. Added to the selector's own score, the gain is indistinguishable from zero on every estimator:

Population	Estimator	ΔAUROC	95% CI
Verified (34 rows, 12 failures)	registered LORO	−0.114	[−0.37, +0.22]
Multilingual (77, 35)	registered LORO	+0.005	[−0.19, +0.22]
Pooled (111, 47)	logistic LORO	−0.030	[−0.10, +0.04]
Pooled	5-fold, clustered	−0.014	[−0.09, +0.07]
Pooled	fixed unit weights	+0.013	[−0.07, +0.12]

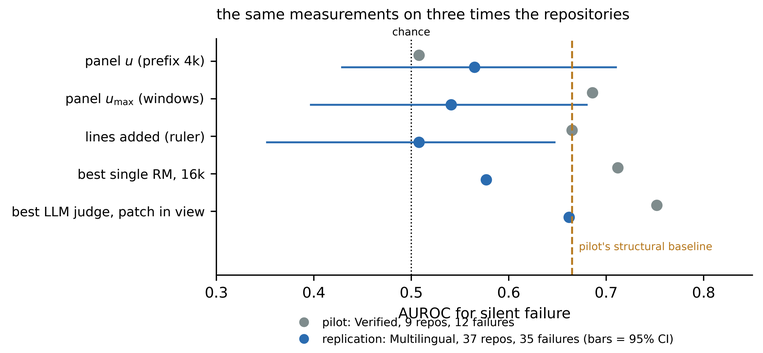
The per-benchmark LORO rows are the registered endpoint. Pooling across benchmarks was added after registration and is descriptive.

+0.12

loosest pooled upper bound on what disagreement adds, over 47 failures and 46 repositories. The bar we declared actionable in advance was +0.05.

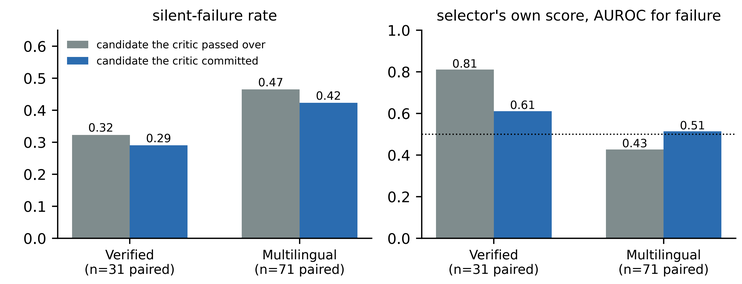
With 47 positives this is a **failure to detect gain**, not evidence of absence; equivalence would need roughly five times the failures. It held across 4k–16k contexts, pruned windows, eight reward models from 1B to 32B, two re-selection regimes and a panel of LLM judges.

The pilot's positives did not travel



On 37 fresh repositories, $u_{\max}$ fell from 0.686 to 0.541 and the 1B critic from 0.712 to 0.485. Every pilot configuration that crossed a bar did so on 12 positives, and none survived the replication.

Why the population matters



On Verified the selector's own score ranks failures at **0.81** among candidates it passed over and **0.61** among the ones it committed, on the same instances. Pool performance does not answer the deployment question.

The critics are fine. Disagreement is the wrong statistic.

`gpt-6-sol` ranks silent failures at **0.776** on the replication and **0.826** on the pilot; the disagreement of that newer pair falls to **0.41**. Test–retest shows the critics are stable ($r \geq 0.98$): the failure is in what disagreement measures, not in critic noise.
Post hoc, at reviewers' request: add `gpt-6-sol`'s *own* score to the selector under the same protocol and the gain is +0.18 [+0.10**,** +0.28] on Multilingual, +0.14 [−0.02, +0.28] on Verified, +0.17 [+0.10, +0.25] pooled (fixed unit weights, LORO). Not pre-registered. The critic's judgment carries; its disagreement with others does not.

Seven ways the instrument lied to us

Each would have produced a publishable result: a sandbox flag that hid `numpy` from the gold harness; a judge emitting three values over 640 candidates; image pulls failing (41/41 "failures" in 499 s); a power function reporting 0.40 under the null; repository grouping that truncated hyphenated names; rank normalisation fit on all rows; a selector-axis filter that manufactured +0.375 where the correct population gives +0.012.
Rule adopted: a rate of exactly 0% or 100%, or a statistic that cannot move, is an instrument reading, not a result.

A stronger agent

Amendment 13: `DeepSeek-V4.1-Flash`, 114 Multilingual instances, 100 steps. Failure rate **0.200** over 456 attempts (four per instance) vs **0.579** for `gpt-5-mini` over 114 (one per instance) at the same 100-step cap; the capability gate passed, but **no critic cleared the 0.65 selector bar** (gpt-5 judge 0.552; reward models 0.419–0.523; gpt-6-sol 0.619), so the cell is descriptive only.